

Reproductive photoresponsiveness in male spiny mice from South Africa

K. Medger¹, C. T. Chimimba^{1,2} & N. C. Bennett¹

1 Department of Zoology and Entomology, Mammal Research Institute (MRI), University of Pretoria, Pretoria, South Africa

2 Department of Zoology and Entomology, DST-NRF Centre of Excellence for Invasion Biology, University of Pretoria, Pretoria, South Africa

Keywords

Acomys spinosissimus; photoperiod; environmental factors; seasonal reproduction; testosterone; gonadal development; subtropics; southern hemisphere.

Correspondence

Katarina Medger, Department of Zoology and Entomology, University of Pretoria, Private Bag X20, Hatfield 0028, South Africa. Tel: +27 12 420 4872; Fax: +27 12 362 5242
Email: kmedger@zoology.up.ac.za

Editor: Andrew Kitchener

Received 31 May 2011; revised 18 September 2011; accepted 30 September 2011

doi:10.1111/j.1469-7998.2011.00872.x

Abstract

Many seasonally breeding mammals use changes in photoperiod as a reliable cue to time reproduction. Photoperiodic timing assists an animal in predicting annual environmental changes in its habitat and therefore, enables it to accurately time reproductive events to the most favourable conditions. Changes in day length are more pronounced in the temperate regions and photoperiod is used as a cue for reproduction by most mammals above 30° latitude; however, a number of subtropical species also use this proximate factor to regulate their reproductive cycle. We investigated the reproductive photoresponsiveness of 14 male spiny mice (*Acomys spinosissimus*) from southern Africa to short-day (SD; 8 h light : 16 h dark) and long-day (LD; 16 h light : 8 h dark) photoperiods. Testicular mass and volume, seminiferous tubule diameter and plasma testosterone concentrations significantly increased in animals subjected to LD and they were regressed when the males were kept under SD. Body mass of the males was not significantly affected by the photoperiodic conditions. Although male *A. spinosissimus* appear to use photoperiod as a proximate factor to regulate reproduction seasonally, other environmental factors, such as rainfall, food quantity and quality as well as temperature, may regulate reproduction in *A. spinosissimus* in concert with photoperiod. In conclusion, the present study demonstrates the significance of photoperiodic time-measuring systems in the regulation of seasonal reproduction in a subtropical rodent.

Introduction

Many mammals occur in habitats where seasonal changes in environmental parameters are predictable throughout the year; however, it is crucial for an animal's survival and reproductive success to be able to anticipate these changes accurately. The absolute day length as well as the direction of day-length change (photoperiod) is used by a large number of mammal species as a proximate cue to time seasonal changes in reproduction as well as other changes in physiology and behaviour (Goldman, 2001). Changes in photoperiod are most pronounced at higher latitudes and it has been proposed that this proximate factor is primarily used by species occurring at latitudes above 30° where the photoperiodic signal is strongest and most reliable (Bradshaw & Holzapfel, 2007). For example, more northerly populations of the white-footed mouse (*Peromyscus leucopus*) and the deer mouse (*P. maniculatus*) were observed to be more reproductively photoresponsive than their southerly populations (Lynch, Heath & Johnston, 1981; Dark *et al.*, 1983). In many small and often short-lived mammals from temperate regions, a continuum of photoresponsive to non-photoresponsive individuals can be found in a single population, which allows for a more plastic

response to environmental changes and enables the non-photoresponsive individuals to breed opportunistically (Prendergast, Kriegsfeld & Nelson, 2001).

In the tropics and subtropics, where photoperiodic changes are less pronounced, most mammals do not use photoperiod as a proximate cue (Bernard & Hall, 1995; Nunes *et al.*, 2002) and some even appear to have abandoned any photoperiodic time-measuring systems (Bronson & Heideman, 1992). However, a few rodent species from subtropical Africa (Muteka, Chimimba & Bennett, 2006) and an Asian shrew species from the subtropics (Wayne & Rissman, 1990) have been found to be reproductively photoresponsive. The climate at the lower latitudes, especially in Africa, is often characterized by the occurrence of one or two rainy seasons, which have a major effect on vegetation growth and therefore, rainfall with (or without) a concomitant increase in food quality and quantity has been suggested to be the main factor influencing seasonal reproduction in mammals throughout most of Africa (Neal, 1986). In the pouched mouse (*Saccostomus campestris*) and the four-striped field mouse (*Rhabdomys pumilio*) from South Africa, it has been demonstrated that food quantity affects reproduction; however, an associated influence of low ambient temperature on reproductive decline was also

suggested (Jackson & Bernard, 2001; Tinney, Bernard & White, 2001).

A large number of other factors, besides photoperiod, rainfall and temperature, such as social cues (Demas & Nelson, 1998), green vegetation (Reichman & van de Graaff, 1975) as well as secondary plant compounds (Sanders *et al.*, 1981; Wube, Haim & Fares, 2009), have been found to directly or indirectly influence reproduction in temperate as well as tropical and subtropical mammal species. It should be noted that none of the factors mentioned earlier is mutually exclusive in the regulation of reproductive function and both proximate as well as ultimate factors may commonly be used in combination to facilitate the best reproductive response to the environmental conditions experienced by a species, population or even single individual (Bronson, 1998).

Reproductive responsiveness to photoperiod has been studied in only a few small mammals from southern Africa, such as *S. campestris* and *R. pumilio* but in both species, photoperiod was found not to be the primary factor regulating reproduction (Bernard & Hall, 1995; Jackson & Bernard, 1999). In contrast, the strongly seasonally reproducing Namaqua rock mouse (*Micaelamys namaquensis* – formerly *Aethomys namaquensis*) and the Tete veld rat (*A. ineptus*) from South Africa have been found to be reproductively photoreponsive (Muteka *et al.*, 2006). In order to gain further insights into the mechanisms shaping seasonality of reproduction in southern African rodents, this study aimed to investigate the reproductive photoresponsiveness in male spiny mice (*Acomys spinosissimus*) from South Africa by comparing the development of the testes and concentrations of plasma testosterone between males subjected to either short-day (SD) or long-day (LD) photoperiods.

A. spinosissimus is relatively widespread in Africa, south of the equator, and in the southern African subregion, it occurs in Mozambique, Botswana and north-eastern South Africa (Skinner & Chimimba, 2005). The habitat of *A. spinosissimus* in South Africa is characterized by one rainy season, which spans from September until April and *A. spinosissimus* has been found to reproduce seasonally (Medger, 2010; Medger, Chimimba & Bennett, 2010). The breeding season was found to coincide with the warm and wet spring and summer months of the southern hemisphere, while breeding ceased during the cold and dry autumn and winter months. It has, therefore, been previously suggested that rainfall, resulting in an increase in food quantity and quality, is the ultimate cause for reproductive seasonality in *A. spinosissimus* (Medger *et al.*, 2010).

We now hypothesize that male *A. spinosissimus* from a population from less than 30° south are reproductively responsive to changing photoperiods because other rodent species that co-occur with *A. spinosissimus*, were also found to be reproductively photoresponsive (see Muteka *et al.*, 2006). In addition, male golden spiny mice (*A. russatus*) were observed to reduce spermatogenesis under SD photoperiods although common spiny mice (*A. cahirinus*) were found to be reproductively non-responsive to photoperiod (Wube, Haim & Fares, 2008). We predicted that the testes of *A. spinosissimus* would regress and the plasma testosterone

concentration would be lower under SD compared with LD photoperiods.

Materials and methods

A total of 14 male *A. spinosissimus* were collected at the end of the breeding season and during the non-breeding season, namely between January and March 2008, and in June 2009. The males were caught in a private game reserve in the Soutpansberg region, Limpopo Province, South Africa (22°58'S, 22°57'S; 29°25'E, 29°24'E) under permits (CPM-333-00002, CPM-002-00002) issued by the Convention for the International Trade of Endangered Species and Permit Management Office, Department of Environmental Affairs, Polokwane, Limpopo Province, South Africa. The animals were caught over-night along the rocky outcrops of the reserve with Sherman live traps (H. B. Sherman Traps, Inc. Tallahassee, FL, USA) baited with a mixture of peanut butter, oats and fish. Male spiny mice were housed in standard polyurethane cages, which were embedded with wood shavings and paper towelling was provided as shelter. They were fed daily with apples and carrots and mouse chow pellets (Die Klein Kooperasie, Pretoria, Gauteng, South Africa) and water was provided *ad libitum* during the entire experiment.

In the laboratory, animals were housed in climate-controlled rooms, which allowed a constant temperature of 25°C throughout the experiment. The males were weighed to the nearest 0.001 g with a digital balance (Scout Pro SPU123, Ohaus Corporation, Pine Brook, NJ, USA) before the onset of the experiments. All males were initially subjected to a photoperiod of 8 h of light and 16 h of darkness (8L : 16D; SD) for 40 days to attain a similar reproductive status for all the males before the start of the actual experimental treatments. Subsequently, male spiny mice were subjected to either a photoperiod of 8L : 16D or 16 h of light and 8 h of darkness (16L : 8D; LD). Seven males were randomly assigned to each experimental treatment. After 30 days on either the LD or SD treatments, the male spiny mice were weighed again and subsequently euthanized with an overdose of halothane. Blood was taken from the heart of all the males by exsanguination and then centrifuged at 500×g for 15 min. The blood plasma was separated from the blood cells and frozen at –35°C until analysis for testosterone concentration. The testes were dissected out and fixed in Bouin's fluid for approximately 20 h, after which they were stored in 70% ethanol. Seminal vesicles were dissected out and immediately weighed to the nearest 0.001 g. All experimental procedures were approved by the animal ethics committee of the University of Pretoria (ethics clearance number: A003-07).

Histology

All excess tissues were removed from the fixed testes before they were weighed separately to the nearest 0.0001 g using a high precision scale (Ohaus Corp., Pine Brook, NY, USA). The testes length and width (mm) were measured with a pair of digital callipers (Sylvac Opto RS 232, Ultra Praezision

Messzeuge GmbH, Glattbach, Germany) to the nearest 0.01 mm and then utilized to calculate testicular volume (mm^3) by using the formula for the volume of an ellipsoid: $V = 4/3 \pi ab^2$, where a represents half the maximum length and b half the maximum width (Woodall & Skinner, 1989). The average of mass (mg) and volume were calculated for both testes per male. The tissues were dehydrated by a series of ethanol baths of increasing concentrations before being embedded in paraffin wax. The testes were then serially sectioned in 7- μm thick sections with a rotary microtome (820 Spencer, American Optical, Scientific Instrument Division, Buffalo, NY, USA) and mounted on microscope slides with gelatine. After approximately 48 h of drying in an oven, the sections were sequentially stained in Ehrlich's haematoxylin and eosin as described by Drury & Wallington (1967). Round seminiferous tubules were photographed at $\times 10$ magnification with a digital camera (Moticam 1000 1.3M Pixel USB 2.0, Motic China Group, LTD., Xiamen, China) attached to a light microscope (Diaplan, Ernst Leitz Wetzlar GmbH). Subsequently, the diameters of 50 randomly selected seminiferous tubules (μm) per testis per animal were measured with the computer programme Motic Images Plus 2.0ML (Motic China Group, LTD., Xiamen, China) and the average of all 100 diameters per individual was calculated.

Testosterone analysis

Plasma testosterone concentrations were measured for all males with a coat-a-count hormone kit (Siemens Medical Solutions Diagnostics, Los Angeles, CA, USA) according to the guidelines provided by the manufacturer. The assay was validated for *A. spinosissimus* by comparing a serial dilution curve with the calibration curve using an analysis of covariance (ANCOVA) and no significant difference between the two curves ($F_{1,3} = 5.87$; $n = 3$; $P = 0.09$) was found. The intra-assay coefficients of variation were 3.5% and 7.8% and the inter-assay coefficient was 5.0%. The minimum detectable amount of testosterone for the assay was 1.39 nmol L^{-1} .

Data analysis

A repeated measures analysis of variance was performed to compare body mass before and after the experiment during which photoperiodic treatment (LD and SD) was used as an in-between factor. ANCOVAs, with body mass measured at the end of the experiment as a covariate, were carried out to compare testicular mass and volume, seminiferous tubule diameter and plasma testosterone concentration between the two photoperiodic treatments. The analysis of plasma testosterone concentration was based on log-transformed data because the original data were non-parametric. Statistical Package for the Social Sciences statistics version 17.0 (SPSS Inc., Chicago, IL, USA) was used for all statistical analyses. All values are given as mean ± 1 standard deviation and results were found to be significant at $P < 0.05$.

Results

Photoperiod affected reproductive development in male *A. spinosissimus*. Seminal vesicles were enlarged in males that

were subjected to LD ($0.11 \pm 0.06 \text{ g}$; range: 0.04–0.24 g) and reduced under SD conditions where seminal vesicles were found to be too small to be weighed. There was no relationship between body mass and testicular mass, seminiferous tubule diameter or plasma testosterone concentration ($F_{1,11} < 4.53$; $P > 0.06$). Testicular volume was, however, significantly positively correlated with body mass ($F_{1,11} > 7.38$; $P < 0.02$). Testicular volume and mass were significantly larger in male spiny mice subjected to LD compared with SD photoperiods ($F_{1,11} > 12.73$; $P < 0.01$; Fig. 1a and b). The diameter of the seminiferous tubules was also significantly larger in males under LD than in males under SD ($F_{1,11} = 13.36$; $P < 0.01$; Fig. 1c) and significantly more plasma testosterone was recorded under LD compared with SD conditions ($F_{1,11} = 31.20$; $P < 0.001$; Fig. 1d).

Male spiny mice were significantly heavier at the end of the experiment ($21.8 \pm 3.5 \text{ g}$) than they were at the beginning ($18.4 \pm 2.5 \text{ g}$; $F_{1,12} = 28.14$; $P < 0.001$). However, we did not detect an influence of photoperiod on male body mass because body mass was not significantly different between SD and LD photoperiods ($F_{1,12} = 0.57$; $P = 0.47$) and also, there was no significant relationship between the photoperiodic treatments (SD and LD) and body mass measured at either the beginning or end of the experiment ($F_{1,12} = 3.02$; $P = 0.11$; Fig. 2).

Discussion

In a previous study, it was demonstrated that *A. spinosissimus* is a seasonal breeder that reproduces during the warm and wet spring and summer months in the southern African sub-region but the factors that lead to this reproductive pattern were unknown (Medger, 2010; Medger *et al.*, 2010). The present study provides the first evidence that male *A. spinosissimus* are reproductively responsive to a change in photoperiod, which may be used as a proximate factor to regulate seasonal reproduction in this species. Male *A. spinosissimus* responded to LD photoperiods with an increase in testicular mass and volume, seminiferous tubule diameter, mass of accessory glands (seminal vesicles) and testosterone concentrations in comparison with SD photoperiods. It is interesting to note that testes size and seminiferous tubule diameter of males subjected to LD photoperiods were similar to those previously found for wild-caught males in July (the start of breeding season in male *A. spinosissimus*; Medger, 2010) and not the size observed for males collected at the peak of the breeding season in September (e.g. testicular volume, LD: $99.6 \pm 23.6 \text{ mm}^3$, July: $100.0 \pm 50.9 \text{ mm}^3$, September: $230.6 \pm 44.2 \text{ mm}^3$; Medger, 2010). This may indicate that 30 days of LD conditions is an insufficient time for maximum growth of the testes and it may be speculated that males kept longer under LD conditions would show a larger increase in testes size and seminiferous tubule diameter than presently reported.

Our findings on the reproductive responsiveness of male *A. spinosissimus* are similar to those of Muteka *et al.* (2006) on the sympatric *M. namaquensis* and *A. ineptus* from South Africa. Both species showed significantly higher testicular mass and volume and larger seminiferous tubule diameters under LD than under SD photoperiodic conditions; however,

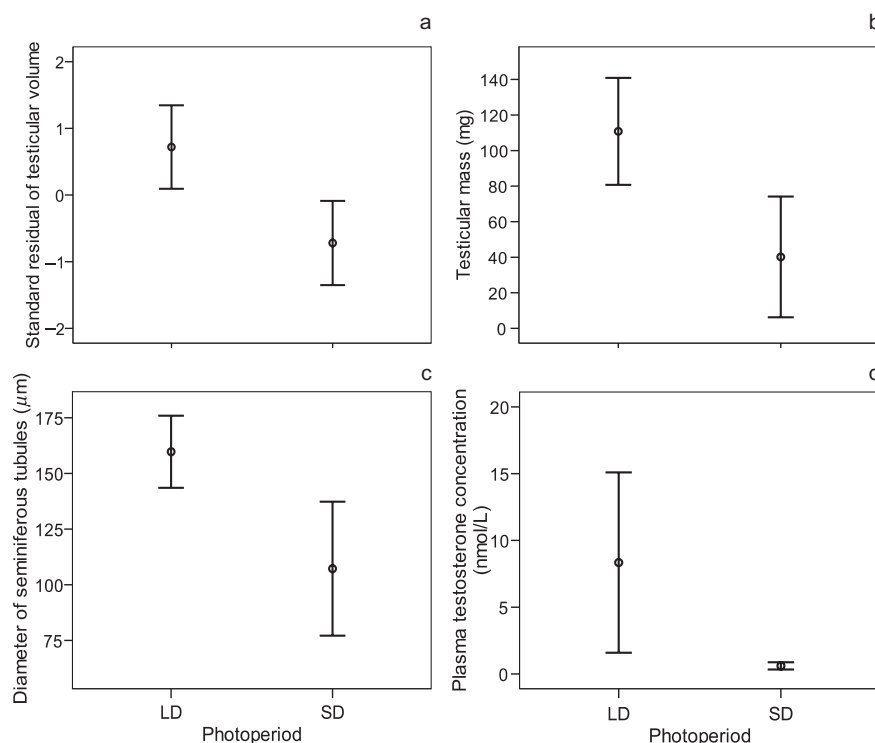


Figure 1 Standardized residual of testicular volume (mm³) by body mass (g; a), testicular mass (mg; b), seminiferous tubule diameter (μm; c) and plasma testosterone concentration (nmol/L; d) of male spiny mice (*Acomys spinosissimus*) from South Africa subjected to either a photoperiod of 16 h light and 8 h darkness (LD; *n* = 7) or 8 h light and 16 h darkness (SD; *n* = 7). Values are presented as mean ± 1 standard deviation.

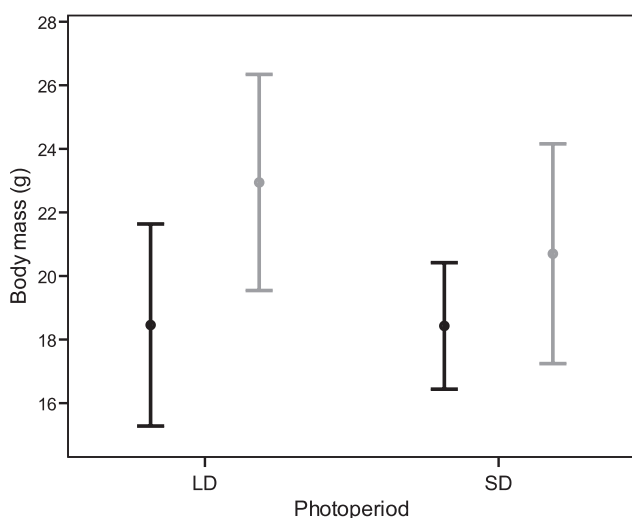


Figure 2 Mean body mass (g) ± 1 standard deviation of male spiny mice (*Acomys spinosissimus*) from South Africa compared between long-day (LD; *n* = 7) and short-day (SD; *n* = 7) photoperiodic treatments and measured at the start (black bars) and end (grey bars) of the experiment.

plasma testosterone concentration was not different between the photoperiodic conditions in *A. ineptus*, although it was higher under LD than SD in *M. namaquensis*. In contrast, day length did not affect either testes size or spermatogenesis in both *S. campestris* and *R. pumilio*, both of which seem to breed opportunistically in southern Africa (Bernard & Hall, 1995; Jackson & Bernard, 1999, 2006). In addition, Nunes *et al.* (2002) found that photoperiod does not influence plasma testosterone concentration, testes size and seminal vesicle mass in males of the seasonally breeding Nile grass rat (*Arvicanthis niloticus*) from an equatorial population. Other rodents, bats and shrews, which occur near the equator, have also been found to be non-responsive to photoperiod (Rissman *et al.*, 1987; Heideman & Bronson, 1990; O'Brien, Curlewis & Martin, 1993). These findings suggest that opportunistically breeding small mammals from tropical and subtropical regions do not use photoperiodic cues to regulate reproduction. However, subtropical rodents, which reproduce seasonally, appear to depend considerably on photoperiod as a proximate factor to anticipate environmental changes and shape reproduction as appears to be the case in *A. spinosissimus*, too. However, the day lengths chosen for the present study were much longer/shorter than the day lengths *A. spinosissimus* would experience in its natural environment. It should, therefore, be noted that the present study tested the reproductive responsiveness to photoperiod, but not the

actual responses, which may be observed in the habitat of *A. spinosissimus*. Future studies should also test if *A. spinosissimus* is also responsive to photoperiods typical for populations near the equator. We, however, propose that male *A. spinosissimus* from the subtropics are able to respond to these photoperiods and, therefore, also use photoperiod to time reproduction in their natural habitat because an exclusion of any photoperiodic time-measuring systems would be expected otherwise.

Responses to photoperiod have been studied in a number of other *Acomys* species. Males of *A. russatus*, for example, were spermatogenically more active when kept under LD photoperiods than SD photoperiods, but females of the same species did not show any reproductive response to photoperiod (Wube *et al.*, 2008). Both male and female *A. cahirinus* were found to be reproductively non-responsive to changing photoperiods (Wube *et al.*, 2008). In another study, however, spermatogenic activity was found to be decreased in *A. cahirinus* subjected to SD photoperiods (El-Bakry, Zahran & Bartness, 1998). Photoperiod and other factors that influence seasonal reproduction have not been investigated in any other *Acomys* species. However, the Cape spiny mouse (*A. subspinosus*) from the Eastern Cape Province of South Africa may be reproductively non-photoresponsive because it is an opportunistic breeder (Fleming & Nicolson, 2002). Latitude does not seem to play a role in the responsiveness to changing day length within the genus *Acomys* and other environmental factors may play a more important role for seasonal reproduction in this genus. Trainor *et al.* (2006) found no distinct relationship between latitude of origin and photoperiodic responsiveness in five species of *Peromyscus* from different latitudes. They, therefore, suggested that, among closely related species, photoperiodic effects on reproductive function may be mediated by different physiological mechanism (Trainor *et al.*, 2006).

Body mass increased during the course of the experiment, which is likely a result of captivity as the animals were provided with a more protein-rich food source and were likely to be less active in the laboratory than in their natural environment. However, there was no effect of photoperiodic treatment on body mass and the males weighed the same under SD and LD photoperiods. It appears to be fairly common in rodents that body mass is not affected by different photoperiods. El-Bakry *et al.* (1998) observed no change in body mass with varying photoperiods in four desert rodents including *A. cahirinus*. In contrast, factors that indicate body condition were affected by photoperiod and for example, fat pad mass and carcass lipid content increased under SD photoperiodic exposure in male *A. cahirinus* (El-Bakry *et al.*, 1998). Although photoperiod appears to be used to time reproduction, it does not seem to be important in regulating body mass in male *A. spinosissimus*. However, moderate photoperiodic effects on body mass may have been masked by the high food availability in the laboratory and body condition of *A. spinosissimus* may be influenced by changing day length.

A. spinosissimus occurs in a highly seasonal habitat where the abundance of high quality food is mediated by seasonal rainfall. Besides photoperiod, food quantity and quality may

regulate reproduction in the spiny mouse as described for other rodents. For example, food deprivation negatively affected reproductive organ growth in deer mice although not in house mice (*Mus musculus*) (Blank & Desjardins, 1984). Protein content of the diet of mammals fluctuates with seasons and it has been suggested that it may limit reproduction in tropical rodents (Field, 1975). In addition, the salt content of plants increases during the dry season in xeric environments because of evaporative water loss and thus, Wube *et al.* (2009) suggested that dietary salinity may be used as a proximate factor to predict the best time for reproduction in *A. russatus*. In addition, low sodium and calcium content of seeds was found to cause lower reproductive success in the California vole (Batzli, 1986). The availability of food often interacts with photoperiod, and food restriction has been found to enhance the effects of SD photoperiods (Nelson *et al.*, 1997) whereas availability of green food may counteract the inhibitory actions of SD (Nelson, Dark & Zucker, 1983).

The natural environment of an animal is much more complex than the conditions provided in laboratory experiments (Bronson, 1998), which may explain some of the discrepancies between the present laboratory study and results on wild-caught animals. If the present results were compared with the results on wild-caught animals (Medger, 2010; Medger *et al.*, 2010), it appears that food quantity and possibly quality may interact with photoperiod to shape reproduction in *A. spinosissimus*. Testicular mass and volume and seminiferous tubule diameters of male *A. spinosissimus* were higher in males under SD (e.g. volume: 34.0 ± 28.0 mm³) than in wild-caught individuals from the non-breeding season [e.g. volume (March): 12.8 ± 8.1 mm³; Medger 2010], which may have been the result of the large amount of high quality food in the laboratory (for effect on body mass see earlier). Moreover, fat pads around the testes appeared to be larger in individuals housed longer in captivity although time in captivity did not seem to have a significant effect on testes mass and volume, seminiferous tubule diameter and testosterone concentration (these non-significant results were not presented in the present study). As it is unlikely that 70 days under SD was too short for total regression of the testes and all males were caught during the non-breeding season when testes are regressed, we suggest that the higher food quantity and quality in the laboratory caused the difference of testicular size between laboratory and wild-caught animals. In addition, ambient temperature may also have had an effect on the size of the testes as ambient temperature was higher during the entire experiment (25°C) than in the natural habitat during winter (<20°C). Nelson *et al.* (1989) demonstrated that cold ambient temperatures further suppress gonadal size in male prairie voles under SD.

Furthermore, other factors may influence reproduction in female *A. spinosissimus*. It is unknown if female *A. spinosissimus* are reproductively photoresponsive and because reproduction of females of both *A. cahirinus* and *A. russatus* have been found to be independent of photoperiodic changes (Wube *et al.*, 2008), reproduction of female *A. spinosissimus* may not be influenced by photoperiod either. The onset of reproduction in female *A. spinosissimus* was previously found

to coincide with the start of the rainy season (Medger *et al.*, 2010). Either rainfall or water availability in general may, therefore, be proximate but very likely ultimate factors regulating seasonal reproduction in female *A. spinosissimus*. In the California mouse (*Peromyscus californicus*), for example, water availability may regulate reproduction independent of either photoperiod or the availability of food (Nelson, Gubernick & Blom, 1995).

In conclusion, male *A. spinosissimus* are reproductively responsive to photoperiod with LD lengths stimulating gonadal development. As a result, photoperiod may be the proximate factor, which regulates seasonal reproduction in males of this species. A number of other environmental factors are also discussed and it is likely that either rainfall by, for example, influencing plant growth and composition, or temperature affect reproduction in *A. spinosissimus*. In addition, an interaction of several factors may be likely. The present results demonstrate that photoperiod may possibly be an important factor in the regulation of seasonal reproduction not only in temperate but also in subtropical and perhaps even tropical rodents. Furthermore, the present study emphasizes the ecological importance for the precise and premature timing of reproductive events in seasonally breeding rodents from the subtropics.

Acknowledgements

This research was funded by a South African Research Chair of Mammalian Behavioural Ecology and Physiology awarded to N.C. Bennett by the South African Department of Science and Technology (DST) and the South African National Research Foundation (NRF). K. Medger acknowledges a doctoral grant from the NRF. We thank the management and staff of the Goro Game Reserve for permission to collect animals, particularly D. Dewsnap is thanked for his support during this research. A. Prins and D. Swanepoel are thanked for help during the field work and for collecting some of the animals.

References

- Batzli, G.O. (1986). Nutritional ecology of the California vole: effects of food quality on reproduction. *Ecology* **67**, 406–412.
- Bernard, R.T.F. & Hall, J. (1995). Failure of the estrous cycle and spermatogenesis to respond to day length in a subtropical African rodent, the pouched mouse (*Saccostomus campestris*). *Biol. Reprod.* **52**, 1291–1295.
- Blank, J.L. & Desjardins, C. (1984). Spermatogenesis is modified by food intake in mice. *Biol. Reprod.* **30**, 410–415.
- Bradshaw, W.E. & Holzapfel, C.M. (2007). Evolution of animal photoperiodism. *Annu. Rev. Ecol. Evol. Syst.* **38**, 1–25.
- Bronson, F.H. (1998). Energy balance and ovulation: small cages versus natural habitats. *Reprod. Fertil. Dev.* **10**, 127–137.
- Bronson, F.H. & Heideman, P.D. (1992). Lack of reproductive photoresponsiveness and correlative failure to respond to melatonin in a tropical rodent, the cane mouse. *Biol. Reprod.* **46**, 246–250.
- Dark, J., Johnston, P.G., Healy, M. & Zucker, I. (1983). Latitude of origin influences photoperiodic control of reproduction of deer mice (*Peromyscus maniculatus*). *Biol. Reprod.* **28**, 213–220.
- Demas, G.E. & Nelson, R.J. (1998). Social, but not photoperiodic, influences on reproductive function in male *Peromyscus aztecus*. *Biol. Reprod.* **58**, 385–389.
- Drury, R.A.B. & Wallington, E.A. (1967). *Carleton's histological technique*. Oxford: Oxford University Press.
- El-Bakry, H.A., Zahran, W.M. & Bartness, T.J. (1998). Photoperiodic responses of four wild-trapped desert rodent species. *Am. J. Physiol.* **275**, 2012–2022.
- Field, A.C. (1975). Seasonal changes in reproduction, diet and body composition of two equatorial rodents. *East Afr. Wildl. J.* **13**, 221–235.
- Fleming, P.A. & Nicolson, S.W. (2002). Opportunistic breeding in the Cape spiny mouse (*Acomys subspinosus*). *Afr. Zool.* **37**, 101–105.
- Goldman, B.D. (2001). Mammalian photoperiodic system: formal properties and neuroendocrine mechanisms of photoperiodic time measurement. *J. Biol. Rhythms* **16**, 283–301.
- Heideman, P.D. & Bronson, F.H. (1990). Photoperiod, melatonin secretion, and sexual maturation in a tropical rodent. *Biol. Reprod.* **43**, 745–750.
- Jackson, C. & Bernard, R.T.F. (1999). Short day length alone does not inhibit spermatogenesis in the seasonally breeding four-striped field mouse (*Rhabdomys pumilio*). *Biol. Reprod.* **60**, 1329–1323.
- Jackson, C. & Bernard, R.T.F. (2001). Gender differences in the inhibitory effect of a reduction in ambient temperature and a reduction in food quality on reproduction in the southern African rodent, *Rhabdomys pumilio*. *Reproduction* **122**, 385–395.
- Jackson, C. & Bernard, R.T.F. (2006). Variation in the timing of reproduction of the four-striped field mouse, *Rhabdomys pumilio*, in the Eastern Cape Province, South Africa. *Afr. Zool.* **41**, 301–304.
- Lynch, G.R., Heath, H.W. & Johnston, C.M. (1981). Effect of geographical origin on the photoperiodic control of reproduction in the white-footed mouse, *Peromyscus leucopus*. *Biol. Reprod.* **25**, 475–480.
- Medger, K. (2010). *The reproductive biology of two small southern African mammals, the spiny mouse, Acomys spinosissimus (Rodentia: Muridae) and the Eastern rock elephant-shrew, Elephantulus myurus (Macroscelidea: Macroscelididae)*. PhD thesis, Pretoria: University of Pretoria.
- Medger, K., Chimimba, C.T. & Bennett, N.C. (2010). Seasonal reproduction in the female spiny mouse from South Africa. *J. Zool.* **282**, 163–170.
- Muteka, S.P., Chimimba, C.T. & Bennett, N.C. (2006). Reproductive photoresponsiveness in *Aethomys ineptus* and

- A. namaquensis* (Rodentia: Muridae) from southern Africa. *J. Zool.* **268**, 225–231.
- Neal, B.R. (1986). Reproductive characteristics of African small mammals. *Cimbebasia* **8**, 113–127.
- Nelson, R.J., Dark, J. & Zucker, I. (1983). Influence of photoperiod, nutrition and water availability on reproduction of male California voles (*Microtus californicus*). *J. Reprod. Fertil.* **69**, 473–477.
- Nelson, R.J., Frank, D., Smale, L. & Willoughby, S.B. (1989). Photoperiod and temperature affect reproductive and non-reproductive functions in male prairie voles (*Microtus ochrogaster*). *Biol. Reprod.* **40**, 481–485.
- Nelson, R.J., Gubernick, D.J. & Blom, J.M.C. (1995). Influence of photoperiod, green food, and water availability on reproduction in male California mice (*Peromyscus californicus*). *Physiol. Behav.* **57**, 1175–1180.
- Nelson, R.J., Marinovic, A.C., Moffatt, C.A., Kriegsfeld, L.J. & Kim, S. (1997). The effects of photoperiod and food intake on reproductive development in male deer mice (*Peromyscus maniculatus*). *Physiol. Behav.* **62**, 945–950.
- Nunes, S., McElhinny, T.L., Mahoney, M.M. & Smale, L. (2002). Effects of photoperiod on the reproductive condition of Nile grass rats (*Arvicanthis niloticus*) from an equatorial population. *Afr. J. Ecol.* **40**, 295–302.
- O'Brien, G.M., Curlewis, J.D. & Martin, L. (1993). Effect of photoperiod on the annual cycle of testis growth in a tropical mammal, the little red flying fox, *Pteropus scapulatus*. *J. Reprod. Fertil.* **98**, 121–127.
- Prendergast, B.J., Kriegsfeld, L.J. & Nelson, R.J. (2001). Photoperiodic polyphenisms in rodents: neuroendocrine mechanisms, costs, and functions. *Q. Rev. Biol.* **76**, 293–325.
- Reichman, O.J. & van de Graaff, K.M. (1975). Association between ingestion of green vegetation and desert rodent reproduction. *J. Mammal.* **56**, 503–506.
- Rissman, E.F., Nelson, R.J., Blank, J.L. & Bronson, F.H. (1987). Reproductive response of a tropical mammal, the musk shrew (*Suncus murinus*), to photoperiod. *J. Reprod. Fertil.* **81**, 563–566.
- Sanders, E.H., Gardner, P.D., Berger, P.J. & Negus, N.C. (1981). 6-Methoxybenzoxazolinone: a plant derivative that stimulated reproduction in *Microtus montanus*. *Science* **214**, 67–69.
- Skinner, J.D. & Chimimba, C.T. (2005). *The mammals of the southern African subregion*. 3rd edn. Cambridge: Cambridge University Press.
- Tinney, G.M., Bernard, R.T.F. & White, R.M. (2001). Influences of food quality and quantity on the male reproductive organs of a seasonally breeding rodent, the pouched mouse (*Saccostomus campestris*), from a seasonal but unpredictable environment. *Afr. Zool.* **36**, 23–30.
- Trainor, B.C., Martin, L.B., Greiwe, K.M., Kuhlman, J.R. & Nelson, R.J. (2006). Social and photoperiod effects on reproduction in five species of *Peromyscus*. *Gen. Comp. Endocrinol.* **148**, 252–259.
- Wayne, N.L. & Rissman, E.F. (1990). Effects of photoperiod and social variables on reproduction and growth in the male musk shrew (*Suncus murinus*). *J. Reprod. Fertil.* **89**, 707–715.
- Woodall, P.F. & Skinner, J.D. (1989). Seasonality of reproduction in male rock elephant shrews, *Elephantulus myurus*. *J. Zool.* **217**, 203–212.
- Wube, T., Haim, A. & Fares, F. (2008). Reproductive response of xeric and mesic populations of the spiny mouse *Acomys* to photoperiod acclimation. *J. Arid Environ.* **72**, 440–447.
- Wube, T., Haim, A. & Fares, F. (2009). Effect of increased dietary salinity on the reproductive status and energy intake of xeric and mesic populations of the spiny mouse, *Acomys*. *Physiol. Behav.* **96**, 122–127.